

The radical as a forced structure : how to split powers from radicals

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Abstract: The n -th root is commonly regarded as a value of the power family: $x^{1/n} = x^t$ at $t = 1/n$. On the positive reals the collapse is correct: there $p_n: x \mapsto x^n$ is an automorphism and its inverse is single-valued. We show the collapse is a torsion phenomenon: it holds precisely where the n -th roots of unity are trivial. Over every commutative group the kernel of the power map is the subgroup $\mu_n = \{x: x^n = 1\}$, every fiber is the coset $x \cdot \mu_n$, and the radical is not a function but a section of the n -fold cover: a choice among n branches differing by multiplication with μ_n . The collapse criterion — p_n is injective if and only if μ_n is trivial — marks where the identification of roots with fractional powers survives. Three elementary theorems express this split (kernel, fibers, collapse criterion) and are machine-verified in Lean 4 with Mathlib: zero proof gaps. Computationally we give certified radical values and a quadratic solver: $x^2 - bx - c = 0$ has a real root if and only if $b^2 + 4c \geq 0$, and both radical roots are exhibited and the polynomial factors. It climbs to the cubic: the Cardano construction exhibits a real root of $x^3 + px + q = 0$ whenever $(q/2)^2 + (p/3)^3 \geq 0$, resting on surjectivity of the cube map on $\mathbb{R}$ — the intermediate value theorem. It then widens instead of climbing: $\sqrt[n]{x} = \exp(\ln x/n)$ is the unique positive solution of $y^n = x$ for every natural degree, so quintic roots and roots of every degree are one statement, not a ladder. Degree five withholds not the pure root but the mixed tower; solvability of the general quintic by radicals is cited (Abel–Ruffini), not proved here. Every statement of the main deduction is machine-checked, and the boundary between executable integer arithmetic and certified, non-executable real statements is explicit: reductions over $\mathbb{R}$ are certified theorems, not computed numeric iterations.

Keywords: radical function, power map, roots of unity, torsion, quotient and section, formal verification

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“Eternity is a long time.”

— Isaac Asimov, Foundation

1. Question, inputs, and logical order

For $t \in \mathbb{R}$ and $x > 0$ the power $x^t = \exp(t \ln x)$ [1] is a function of two arguments, single-valued, and nothing prevents evaluating it at $t = 1/n$. This yields the received convention

$$\text{“the } n\text{-th root is the fractional power } x^{1/n}, \text{ i.e. } x^t \text{ at } t = 1/n\text{”}. \quad (1)$$

On the positive reals the convention looks harmless, and is harmless : the power map $p_n: x \mapsto x^n$ is strictly increasing, hence injective ; as a group endomorphism of $\mathbb{R}_+$ it is an automorphism, and its inverse is exactly the value of (1). No choice is ever made, because $\exp$ and $\ln$ are functions, not relations.

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The convention answers, before it is asked, the question of how to invert the map $x \mapsto x^n$. This paper asks where that inversion fails, and which structure the failure forces into existence. Over a general multiplicative structure p_n is not injective, and for a group homomorphism injectivity is equivalent to a trivial kernel. The kernel of p_n is the subgroup of n -th roots of unity, $\mu_n = \{x : x^n = 1\}$; over $\mathbb{C}^\times$ it is a cyclic group of order n [2]. The failure is not a defect of one formula; it is measured by μ_n . Once μ_n is present, the fiber of p_n over x^n is the whole coset $x \cdot \mu_n$, and the symbol $x^{1/n}$, if it names anything, names a choice among n preimages. Inverting powers therefore forces three objects into existence : the subgroup μ_n that is the kernel, the cosets that are the fibers, and the sections — the branch choices — that textbooks write as roots. The radical is a section of an n -fold cover, not a function.

The main results can be stated without machinery, and we state them now ; proofs follow in the sections indicated.

- *The torsion split.* For every commutative group G and every n , the kernel of p_n is exactly $\mu_n(G)$ (Theorem 3), every fiber of p_n is the coset $x \cdot \mu_n(G)$ (Theorem 4), and p_n is injective if and only if $\mu_n(G)$ is trivial (Theorem 7). The identification (1) survives exactly where the last criterion holds ; over $\mathbb{R}_+$ it holds because $\mathbb{R}_+$ is torsion-free. All three theorems are machine-verified in Lean 4 with Mathlib, zero proof gaps.
- *A certified solver ladder.* Over $\mathbb{R}$ the programme is certified inside the proof system. For real b, c , the equation $x^2 - bx - c = 0$ has a real root if and only if $\Delta = b^2 + 4c \geq 0$; when it does, both radical roots are exhibited and the polynomial factors (Theorem 9). The ladder climbs to the cubic : whenever $(q/2)^2 + (p/3)^3 \geq 0$ the Cardano construction exhibits a real root of $x^3 + px + q = 0$, with surjectivity of the cube map as its only analytic input (Theorem 10). The ladder then stops climbing and widens : one statement, $\sqrt[n]{x} = \exp(\ln x/n)$, is the unique positive solution of $y^n = x$ for every degree n at once (Theorem 11). There is no separate quintic theorem.
- *Where the wall is.* Pure n -th roots exist for every degree. What withholds at degree five is the mixed tower : a general quintic is not solvable by field operations and extraction of roots of arbitrary degree. That statement (Abel–Ruffini [3], in the Galois formulation [4]) is cited as the boundary of this paper, not proved here ; its formalization is not claimed.

The strategy is uniform. On $\mathbb{R}_+$ every power and radical is constructed from $\ln$ and $\exp$, and there p_n is an automorphism, so (1) holds verbatim. Over an arbitrary commutative group no analytic tools exist, and the proofs reduce to the ring identity $(xy)^n = x^n y^n$ and its consequences : kernel, fibers, collapse criterion. The certified solvers split the same way. After the substitution $(\sqrt{\Delta})^2 = \Delta$ the quadratic is pure algebra, factorizing $x^2 - bx - c = (x - r_+)(x - r_-)$, and the negative-discriminant case is completing the square ; the cubic needs exactly one analytic input, surjectivity of the cube map on $\mathbb{R}$ — Mathlib’s intermediate value theorem — after which the Cardano identity is again pure algebra. The entire deduction contains one intermediate-value argument.

A boundary is kept explicit throughout, between what is proved inside the proof system and what is cited. Inside : the three structure theorems, the radical examples, and the quadratic, cubic, and every-degree theorems are formalized in Lean 4 [5] with Mathlib [6], in the module `Nods.Theories.Radical` of the NODS repository ; the module is in the main build, and a search over the source tree confirms zero sorry. Cited, not re-proved : the classical context of roots of unity [2], the provenance of the Cardano construction [7], and the Abel–Ruffini and

Galois statements about mixed towers [3, 4]. On the reals, arithmetic is certified, not executed : $\sqrt{}$ is defined by completeness and is not executable, and reductions such as $\sqrt{16} = 4$ via $\sqrt{a^2} = |a|$ are certified theorems, not numeric iteration. We claim certified radical statements, not a numeric radical evaluator.

The order of the paper follows the questions just raised. Section 2 fixes the two objects, the power map and the subgroup of roots of unity, over an arbitrary commutative group. Section 3 records the collapse on $\mathbb{R}_+$ and its failure over $\mathbb{C}^\times$. Section 4 states and proves the three structure theorems — kernel, fibers as cosets, collapse criterion — and reads them as quotient and section. Section 5 draws the concrete consequence for a single number. Section 6 gives the certified quadratic solver, the Cardano cubic, and the radical of every degree, and states the two boundaries (executability, and the degree-five wall against mixed towers). Section 7 records the formalization choices, and Section 8 closes.

1.1. Related work

The literature this paper touches is old and settled; the contribution is not new mathematics but a certified statement of where a convention holds. Roots of unity were studied systematically by Gauss [2]; the impossibility of solving the general quintic by radicals was proved by Abel [3], and Galois turned the obstruction into group theory [4]. Fractional powers and the exponential–logarithm calculus that make $x^{1/n}$ a function on $\mathbb{R}_+$ are Euler’s [1]. The cubic solution certified here goes back to Cardano’s *Ars Magna* [7]. On the formal side, Lean [5] and its mathematical library Mathlib [6] supply the real analysis — exp, ln, $\sqrt{}$, the intermediate value theorem — that the certified statements reuse rather than rebuild.

- Classical radical algebra asks which equations are solvable by adjoining roots; we ask the narrower question, for one power map, what its kernel is and what its sections are, and answer it for every commutative group.
- Euler’s $x^{1/n}$ is correct on $\mathbb{R}_+$ not because of any special property of exp, but because $\mu_n \cap \mathbb{R}_+ = 1$; the analysis here locates the reason in torsion.
- The Cardano construction is four centuries old; Theorem 10 adds certification and makes explicit a third appearance of the torsion split — v cannot be chosen as an independent cube root.
- Mathlib already contains real roots and powers; the contribution of the present module is the splitting statements (kernel, fibers, collapse criterion over arbitrary commutative groups) and the certified solvers, with zero proof gaps.
- Abel–Ruffini and Galois remain cited, not formalized; the boundary between executable integer arithmetic and certified, non-executable real statements is stated, not erased.

2. Setting : power maps and roots of unity

Fix a commutative group G , written multiplicatively, and $n \in \mathbb{N}$. We use two objects.

Definition 1 (Power map). $p_n: G \rightarrow G$ denotes $p_n(x) = x^n$.

Definition 2 (Roots of unity). $\mu_n(G) = \{x \in G : x^n = 1\}$ is the subgroup of n -th roots of unity.

That $\mu_n(G)$ is a subgroup is immediate from commutativity $((xy)^n = x^n y^n)$ and from $(x^{-1})^n = (x^n)^{-1}$. Its elements are the *torsion* of exponent n ; $\mu_n(G) = \{1\}$ is the statement that G has no nontrivial n -th root of unity.

p_n is a group homomorphism : $p_n(xy) = (xy)^n = x^n y^n = p_n(x)p_n(y)$, again by commutativity.

Two base groups. Two instances of the setup run through the paper. On $G = \mathbb{R}_+$, the multiplicative group of positive reals, the equation $x^n = 1$ forces $x = 1$, so $\mu_n(\mathbb{R}_+) = \{1\}$: the roots of unity are trivial, the group is torsion-free. On $G = \mathbb{C}^\times$ the same subgroup is the regular n -gon $\mu_n(\mathbb{C}^\times) = \{e^{2\pi i k/n} : 0 \leq k < n\}$, a cyclic group with n elements [2]. Abstractly, cardinality is never needed : the theorems of Section 4 are stated for an arbitrary commutative G and never count roots of unity. That $\mu_n(\mathbb{C}^\times)$ has exactly n elements is a concrete fact about the complex exponential, not an input to the mechanism ; on $\mathbb{R}_+$ the same subgroup is trivial, and that single fact is the whole difference between the two worlds.

Worked micro-example ($n = 2$). On $G = \mathbb{C}^\times$, $\mu_2 = \{1, -1\}$, and the fiber of p_2 over x^2 is $\{x, -x\}$: both leaves square to x^2 , and nothing in the group prefers one over the other. The square root of x^2 is therefore a two-element choice ; the two values differ by the kernel element -1 . On $G = \mathbb{R}_+$ the contrast is sharp : for $x > 0$ the fiber over x^2 is the singleton $\{x\}$, since $-x \notin \mathbb{R}_+$. The two-element choice collapses to one element, $\sqrt{x^2} = x$ is forced, and the collapse of Section 3 is already visible here in its simplest form.

3. The collapse and its failure

On $G = \mathbb{R}_+$ (positive reals, multiplication), the equation $x^n = 1$ has only the solution 1 ; indeed x^n is strictly increasing. Hence p_n is injective on $\mathbb{R}_+$. It is also surjective : every $y > 0$ is $y = (\exp(\frac{1}{n} \ln y))^n$. Being an injective and surjective group endomorphism, p_n is an automorphism, and every y has exactly one n -th root, the fractional power $y^{1/n} = \exp(\frac{1}{n} \ln y)$. The collapse (1) is exactly the statement that this inverse is *unique*, and uniqueness is the absence of torsion.

Over $\mathbb{C}^\times$ the same equation has n solutions, namely the regular n -gon $\mu_n(\mathbb{C}^\times) = \{e^{2\pi i k/n} : 0 \leq k < n\}$, a cyclic group of order n [2]. The failure of injectivity is measured by this group. The next section states the mechanism once, for every commutative group.

Where does the collapse hide the torsion ? In the branch family of the logarithm. Fix $x > 0$, view $\mathbb{R}_+ \subset \mathbb{C}^\times$, and write $\zeta = e^{2\pi i/n}$. The complex n -th roots of x are

$$\exp\left(\frac{\ln x + 2\pi i k}{n}\right) = \exp\left(\frac{\ln x}{n}\right) \zeta^k = x^{1/n} \zeta^k, \quad k = 0, \dots, n-1,$$

with $\ln$ the real logarithm. The additive branch correction $2\pi i k/n$ inside the exponential becomes, on exponentiation, exactly the kernel element ζ^k outside it : the family already carries the μ_n factor, and it is nothing but the fiber $x^{1/n} \mu_n$ of Theorem 4. The real-positive branch is the single leaf $k = 0$, where $\zeta^k = 1$: there μ_n is invisible, because among the n leaves exactly one — the one whose kernel factor is the identity — lies on the positive axis. The principal branch of the complex logarithm makes the same selection by a cut rather than by the group : it chooses the leaf $k = 0$ discontinuously and keeps the torsion out of sight.

Remark. Writing $x^{1/n}$ never made an equation single-valued ; it only ever recorded that it already was. The collapse holds on $\mathbb{R}_+$ because the group is torsion-free, $\mu_n(\mathbb{R}_+) = \{1\}$, not

because one notation is better than another. Over $\mathbb{C}^\times$ no notational reform can repair it : $\mu_n(\mathbb{C}^\times) \neq \{1\}$, the fibers have several leaves, and no right inverse of p_n can be both continuous (the principal branch is a discontinuous choice) and compatible with the group law (a homomorphic section s would satisfy $s(1) = 1$ and, since $\zeta^n = 1$, also $s(1) = s(\zeta^n) = \zeta$ for any nontrivial $\zeta \in \mu_n(\mathbb{C}^\times)$). Replacing symbols changes nothing, because the obstruction lives in the group, not in the alphabet.

4. Power as quotient, radical as section

Theorem 3 (Kernel is the group of roots of unity). *For every commutative group G and every n , $\ker p_n = \mu_n(G)$.*

Démonstration. An element x lies in the kernel of p_n iff $p_n(x) = 1$ iff $x^n = 1$ iff $x \in \mu_n(G)$. $\square$

What it says. The elements killed by the power map — those with $x^n = 1$ — are precisely the n -th roots of unity : the kernel of p_n is the torsion of exponent n in G . On $\mathbb{R}_+$ this kernel is trivial ; on $\mathbb{C}^\times$ it is the full regular n -gon.

Strategy. Prove equality of subgroups by extensionality : an element lies in $\ker p_n$ exactly when it lies in $\mu_n(G)$, and every step is a definitional equality — unfolding kernel, p_n , and μ_n turns $p_n(x) = 1$ into $x^n = 1$ verbatim. This is why the formal proof is a definitional check : no input beyond the definitions is needed.

Theorem 4 (Fibers are cosets). *For every commutative group G , every n , and all $x, z \in G$,*

$$z^n = x^n \iff \exists \zeta \in G, \zeta^n = 1 \wedge z = x\zeta.$$

Equivalently, the fiber of p_n over x^n is the coset $x \cdot \mu_n(G)$.

Démonstration. ($\Leftarrow$) If $z = x\zeta$ with $\zeta^n = 1$, then $z^n = (x\zeta)^n = x^n \zeta^n = x^n$ by commutativity. ($\Rightarrow$) If $z^n = x^n$, put $\zeta = x^{-1}z$. Then $\zeta^n = (x^{-1}z)^n = (x^n)^{-1}z^n = 1$ and $z = x\zeta$. $\square$

What it says. Two elements are folded to one image by p_n exactly when they lie in one coset of μ_n : equal n -th powers hold if and only if the two elements differ by an n -th root of unity. The fiber over the value x^n is the leaf set $x \cdot \mu_n(G)$ — all preimages of one point — and this is why “the root of x^n ” is a choice among leaves rather than a computation.

Strategy. Two directions, one line each. ($\Leftarrow$) Write $z = x\zeta$ with $\zeta^n = 1$ and expand, using commutativity : $z^n = x^n \zeta^n = x^n$. ($\Rightarrow$) Given $z^n = x^n$, put $\zeta = x^{-1}z$, as in the proof above. Each direction is the definition of p_n plus the homomorphism property.

Corollary 5 (Roots of one value form one coset). *For all $x, z \in G$,*

$$z^n = x^n \iff z \in x \cdot \mu_n(G),$$

i.e. the fiber of p_n over its value at x is the coset $x \cdot \mu_n(G) = \{x\zeta : \zeta^n = 1\}$. In particular, if z is a root of x^n , then so is $z\zeta$ for every $\zeta \in \mu_n(G)$.

Démonstration. Immediate restatement of Theorem 4. $\square$

Corollary 6. *If x^n has one n -th root x_0 , then it has exactly $|\mu_n(G)|$ of them, namely ζx_0 with $\zeta \in \mu_n(G)$. Over a field of characteristic not dividing n that contains a primitive n -th root of unity — for instance $\mathbb{C}$ — this number is exactly n .*

Theorem 7 (Collapse criterion). *For every commutative group G and every n , p_n is injective if and only if $\mu_n(G) = \{1\}$.*

Démonstration. By Theorem 3, $\ker p_n = \mu_n(G)$; a group homomorphism is injective iff its kernel is trivial. $\square$

What it says. The power map is one-to-one exactly when the only n -th root of unity is 1 : injectivity fails precisely to the extent that torsion is present. On $\mathbb{R}_+$ the criterion holds trivially — this is the warrant for the collapse of Section 3 ; over $\mathbb{C}^\times$ it fails, and the single-valued collapse is not available there.

Strategy. For a group homomorphism, injectivity is equivalent to a trivial kernel. Substitute the kernel computed in Theorem 3 : p_n is injective iff $\mu_n(G) = \{1\}$. In the proof system this is the equivalence , applying the standard kernel-trivial-iff-injective lemma to the identification of the kernel of the power map with the roots of unity.

Corollary 8 (The power map is injective on the positive reals). *For every n , $p_n : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is injective : the positive reals have no nontrivial n -th root of unity.*

Démonstration. For $y > 0$, $y^n = 1$ forces $y = 1$ (the map $t \mapsto t^n$ is strictly increasing on $\mathbb{R}_+$) ; hence $\mu_n(\mathbb{R}_+) = \{1\}$, and Theorem 7 applies. $\square$

The content of the three theorems is a division of labour between two structural roles.

- *Power is a quotient.* p_n collapses the orbit of x under multiplication by μ_n to a single point ; it is the projection of G onto G/μ_n (up to the canonical identification). Its kernel is nontrivial exactly when torsion is present.
- *The radical is a section.* A “root” is a choice of one preimage under p_n . Globally this choice cannot be made continuously or canonically ; it exists only branch by branch, and the branches differ by the deck group μ_n . The fractional power $x^{1/n} = e^{(\ln x)/n}$ is the section on the principal branch — the branch on which the torsion is invisible. This is why (1) “works” on $\mathbb{R}_+$ and fails on $\mathbb{C}^\times$: not because the algebra changes, but because the cover is trivial there.

Read the two panels as the two roles of this section at $n = 4$. In the left panel keep your eye on one image value : p_4 folds four distinct leaves into it, so the inverse problem — which point maps here ? — has four answers ; this is the quotient role, the identification of a whole coset. In the right panel keep your eye on one leaf : multiplication by i moves it to the next leaf and back after four steps : the free transitive action of the kernel on the fiber. Between the two panels stands Theorem 4 : the leaves over one image form one coset of μ_4 , and a radical is a choice of one of them.

5. What torsion forces : explaining a number

Two mechanisms are at work in what follows, and it is worth keeping them apart. Whether a fiber is nonempty is an *arithmetic* question : over $\mathbb{Q}$ the equation $x^2 = 2$ has no solution at all

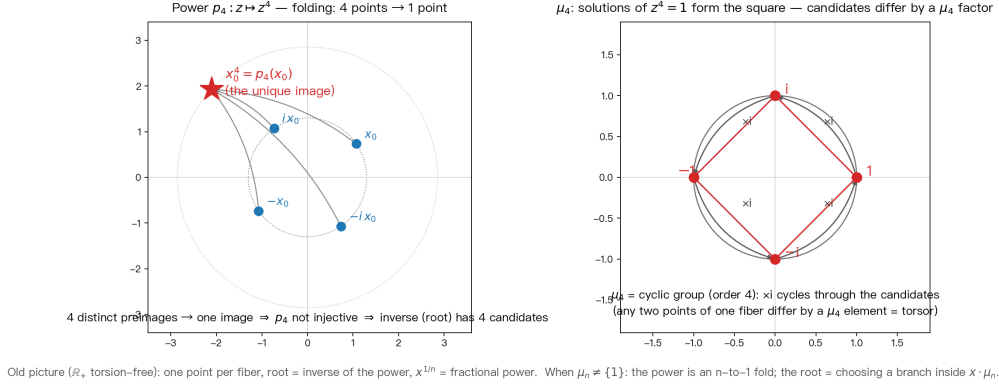
Why “root $\neq$ fractional power” forces torsion μ_n (here μ_4)


FIGURE 1. Left : p_4 folds the four distinct points $x_0, ix_0, -x_0, -ix_0$ of one fiber onto the single image x_0^4 , so the inverse problem “which point maps here?” has four answers, and the radical is a choice among leaves rather than a value. Right : the four leaves are not isolated; they form the coset $x_0\mu_4$ of the kernel $\mu_4 = \{1, i, -1, -i\}$ of p_4 , and multiplication by i cycles through them, returning to the start after four steps. The power map identifies a whole coset to one point, and the kernel acts freely and transitively on the leaves above that point — the torsor structure that a choice of root must break.

(machine-checked below), and the radical is forced into existence as a new object rather than chosen as a symbol. When a fiber is nonempty, its size is a *torsion* question (Corollary 6) : over $\mathbb{R}$ the two roots of $x^2 = 2$ are $\pm\sqrt{2}$, which differ by $\mu_2 = \{\pm 1\}$. Existence is arithmetic; multiplicity is torsion.

- no $n \in \mathbb{N}$ satisfies $n^2 = 2$;
- no $q \in \mathbb{Q}$ satisfies $q^2 = 2$;
- over the reals, $\sqrt{2}$ exists and $(\sqrt{2})^2 = 2$, and the exponential/logarithm path gives the different exact formula $e^{\ln 2} = 2$.

The same number admits *different* exact constructions, and which families succeed depends on the number. Three families run through this paper : pure powers ($25 = 5^2$), radicals ($\sqrt{25} = 5$, an executable integer square root), and exponential–logarithmic formulas ($\exp(\ln 25) = 25$, the identity $\exp \circ \ln = \text{id}$ on $\mathbb{R}_+$). The number 2 separates the first family from the other two : no natural number and no rational number squares to 2, so no power formula explains it, and it is reached only once a radical object is forced into existence. The golden section φ is a radical object in its own right : $\varphi = (1 + \sqrt{5})/2$ solves $x^2 - x - 1 = 0$, and the verification is $\varphi^2 = \varphi + 1$. Table 1 records the exhibit; an em dash marks the column in which the family fails on that number.

	power	radical	exp–log
25	$25 = 5^2$	$\sqrt{25} = 5$	$\exp(\ln 25) = 25$
2	—	$(\sqrt{2})^2 = 2$	$e^{\ln 2} = 2$
φ	—	$\varphi^2 = \varphi + 1$	—

TABLE 1. A number explained by whichever families succeed on it. Entries are certified in the module or are direct instances of its theorems (the identity $\exp \circ \ln = \text{id}$ on $\mathbb{R}_+$); an em dash marks a family that fails on that number.

These are statements, not heuristics : the power claims are ring computations, $\sqrt{25} = 5$ is an executable integer reduction, and the exponential–logarithmic claims are instances of the identity $\exp \circ \ln = \text{id}$. Each row is a different certified route inside the same proof system.

6. Certified radicals and a general quadratic solver

The examples of Section 5 are instances of a general solver. Fix real b, c and the equation

$$x^2 - bx - c = 0, \quad \Delta = b^2 + 4c. \quad (2)$$

Whence the radicals. Completing the square exhibits the two roots before they are named. From $x^2 - bx - c = 0$,

$$x^2 - bx = \left(x - \frac{b}{2}\right)^2 - \frac{b^2}{4} = c,$$

so that

$$(2x - b)^2 = b^2 + 4c = \Delta.$$

Every root therefore satisfies the displayed identity, and conversely any x satisfying it is a root, by reversing the two steps. With $\Delta \geq 0$ the square root $\sqrt{\Delta}$ exists on $\mathbb{R}$, and $2x - b = \pm\sqrt{\Delta}$ yields exactly the two candidates of Theorem 9. The failure direction is visible in the same equation : when $\Delta < 0$, the left side $(2x - b)^2$ is never negative.

Theorem 9 (Quadratic radical solver). *Let $\Delta = b^2 + 4c \geq 0$. Put $r_+ = (b + \sqrt{\Delta})/2$ and $r_- = (b - \sqrt{\Delta})/2$. Then*

1. $r_+^2 - br_+ - c = 0$ and $r_-^2 - br_- - c = 0$;
2. for every x , $x^2 - bx - c = (x - r_+)(x - r_-)$ — there are no other roots ;
3. if $\Delta < 0$ there is no real root at all.

Consequently a real root exists if and only if $\Delta \geq 0$.

Démonstration. All claims are obtained by expanding and using $(\sqrt{\Delta})^2 = \Delta$, which is the defining property of the real square root ; item 1 and item 2 are pure ring identities after that substitution, and item 3 is completing the square : $(2x - b)^2 = \Delta$, impossible for $\Delta < 0$. $\square$

Where the radicals come from. Cardano's substitution is a division of labour between two unknowns. Put $x = u + v$ and impose $3uv = -p$; this is one scalar condition on two degrees of freedom, and the leftover freedom is exactly the μ_3 -torsor of Section 4 : the pairs (u, v) and $(\zeta u, \zeta^{-1}v)$ satisfy the same condition for every cube root of unity ζ . Under the condition the cubic collapses :

$$\begin{aligned} x^3 + px + q &= (u + v)^3 + p(u + v) + q \\ &= u^3 + v^3 + (3uv + p)(u + v) + q = u^3 + v^3 + q, \\ u^3 v^3 &= (uv)^3 = (-p/3)^3 = -p^3/27. \end{aligned}$$

Hence u^3 and v^3 sum to $-q$ and multiply to $-p^3/27$: they are the two roots of the auxiliary quadratic

$$z^2 + qz - \frac{p^3}{27} = 0,$$

whose discriminant is $\Delta_3 = (q/2)^2 + (p/3)^3$. If $\Delta_3 \geq 0$, set $s = \sqrt{\Delta_3}$; the two auxiliary roots are $-q/2 \pm s$, and the theorem below builds a real root out of them. Note that u must be chosen among the three cube roots of $-q/2 + s$ — that freedom is the μ_3 -torsor again — and v is then defined from u by $v = -p/(3u)$, never chosen independently : an independent choice would let uv differ from $-p/3$ by a cube root of unity, and $u + v$ would fail to be a root. Machine verification forces the definitional choice .

Theorem 10 (Cubic radical solution of Cardano). *Let $\Delta_3 = (q/2)^2 + (p/3)^3 \geq 0$. Then $x^3 + px + q = 0$ has a real root, constructed as follows. Put $s = \sqrt{\Delta_3}$, choose u with $u^3 = -q/2 + s$ (possible because the cube map is surjective on $\mathbb{R}$), and set $v = -p/(3u)$. Then $v^3 = -q/2 - s$, $uv = -p/3$, and $u + v$ is a root.*

Démonstration. $u^3 + v^3 = -q$ together with $3uv = -p$ gives $(u + v)^3 = u^3 + v^3 + 3uv(u + v) = -q - p(u + v)$, hence $(u + v)^3 + p(u + v) + q = 0$. The identity $v^3 = -q/2 - s$ follows from $(q/2)^2 - s^2 = -(p/3)^3$ and $u \neq 0$ by direct calculation (the case $u = 0$ forces $p = 0$, reducing to $x^3 + q = 0$). Note that v cannot be chosen as an independent cube root of $-q/2 - s$: then uv would differ by a cube root of unity — the third appearance of the torsion split of Section 4, which machine verification forces us to face. $\square$

Why exp and log here, and only here. The formula $\sqrt[n]{x} = \exp(\ln x/n)$ needs $\ln$ to be a single-valued function and the collapse (1) to hold; both are true on $\mathbb{R}_+$, which is torsion-free ($\mu_n(\mathbb{R}_+) = \{1\}$, Theorem 7). Over $\mathbb{C}^\times$ the same formula would force $\ln$ to choose a branch, and the branch choice is exactly a choice through μ_n : the exp/log route is the section written on the branch where the torsion is invisible. That is why the construction is a statement about $\mathbb{R}_+$, not about formula convenience.

Theorem 11 (Radical of every degree). *Let $n \geq 1$ be any degree and let $x > 0$. On $\mathbb{R}_+$ the power map p_n is injective (Theorem 7 : $\mu_n(\mathbb{R}_+) = \{1\}$), and the unique positive solution y of $y^n = x$ is*

$$y = \sqrt[n]{x} := \exp\left(\frac{\ln x}{n}\right), \quad (3)$$

with the following machine-checked consequences, for every $n \geq 1$:

1. $(\sqrt[n]{x})^n = x$;
2. if $a > 0$ and $a^n = x$ then $a = \sqrt[n]{x}$ — the radical is the inverse of the power map, single-valued;
3. accordingly one formal statement covers every degree : $n = 2, 3, 5, \dots$ at once.

The quintic makes the point : $(\sqrt[5]{32})^5 = 32$ and $\sqrt[5]{25} = 2$ are instances of the same theorem as $\sqrt[3]{27}$ and $\sqrt{9}$; so is $a^7 = 128 \Rightarrow a = \sqrt[7]{128}$. “Roots of arbitrary degree” here means : for every natural degree simultaneously — not a limit of a family of functions as $n \rightarrow \infty$, but one theorem true for all n at once.

Démonstration. By the definition, $(\sqrt[n]{x})^n = \exp(n \cdot \ln x/n) = \exp(\ln x) = x$, which is item 1. For injectivity on $\mathbb{R}_+$, apply $\ln$ to $a^n = b^n$: $n \ln a = n \ln b$, hence $\ln a = \ln b$ (division by $n \neq 0$), hence $a = b$ by exponentiation; item 2 combines this with item 1 and needs no separate positivity hypothesis on x , since $a > 0$ together with $a^n = x$ already forces $x > 0$. $\square$

The statements are machine-checked. Two boundaries are stated explicitly.

- *Executability.* On $\mathbb{N}$ the integer square root $\sqrt{25} = 5$ is computed and reduced by the proof system itself (`norm_num`). On $\mathbb{R}$ the square root is defined by completeness and is not executable; the reductions above (for example $\sqrt{16} = 4$ via $\sqrt{a^2} = |a|$) are certified theorems, not numeric iterations. We do not claim a numeric radical evaluator; we claim certified radical statements.
- *The wall at degree five is a wall against mixed towers, not against roots.* Pure n -th roots exist for every degree (Theorem 11); what dies at degree 5 is the *mixed* radical expression: the general quintic $x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$ is not solvable by any tower of field operations and roots of arbitrary degree (Abel–Ruffini [3]); stating that boundary precisely is the Galois half [4] of the programme and is not claimed here.

7. Formalization notes

All statements marked with a Lean cross-reference above are proved in Lean 4 [5] at version 4.21.0 with Mathlib [6], in the single module `Nods.Theories.Radical` of the NODS repository, imported by the main chain file `Nods.lean`. The proofs are short — Theorem 3 is definitional, Theorem 4 is one calculation in each direction, Theorem 7 is an instance of the standard kernel-trivial-iff-injective lemma — and each engineering choice below records an obstacle met while making those proofs go through.

- *Generic statements.* Every theorem is stated over an arbitrary commutative group G (`CommGroup`) rather than over $\mathbb{R}_+$ alone, so the collapse criterion reads as a theorem about torsion in any group and not as an accident of the positive reals — a choice that costs nothing, since the arguments use only commutativity and the group laws.
- *An in-file kernel subgroup.* The roots of unity are built in-file as `nRoots n G`, with carrier the set of x satisfying $x^n = 1$, rather than imported from Mathlib’s units-based `rootsOfUnity`; that library object lives on the unit group (elements of $G^\times$ shipped with an inverse witness), while the kernel of the power map is a plain subset of G , and coercing between the two copies would buy nothing but transport lemmas at every use.
- *The power map is reused.* The map p_n is not redefined: the theorems use Mathlib’s `powMonoidHom` directly, so homomorphism facts, the kernel lemma, and the subgroup API apply to it without glue.
- *Zero-sorry, zero-warning policy.* The module is held to a strict standard: no sorry occurs in the source tree, the module adds no axiom of its own, and the build is clean — “machine-checked” refers to the checked-in file, not to a transcript of a session.
- *The cube root is proved, not imported.* Surjectivity of the cube map on $\mathbb{R}$ — the analytic primitive behind the Cardano step — is proved in-system from Mathlib’s intermediate value theorem on a bounded interval (`intermediate_value_Icc`), because Mathlib has no cube-root primitive; the IVT proof is the primitive (Theorem 10).
- *Roots of every degree via exp and log.* For arbitrary degree no new primitive is built either: $\sqrt[n]{x}$ is defined as $\exp(\ln x / n)$, and the two claims $(\sqrt[n]{x})^n = x$ and uniqueness are derived from `exp_nat_mul` and `log_pow` (via injectivity of $\ln$ on $\mathbb{R}_+$), so one theorem covers every $n \geq 1$ at once (Theorem 11).
- *Noncomputable reals, executable naturals.* Executability is claimed only where it exists:

`Real.sqrt`, `Real.exp`, `Real.log`, and the degree- n root are noncomputable constants fixed by real analysis, while `Nat.sqrt` is computable; the executed examples are integer ones reduced by `norm_num` ($\sqrt{25} = 5$), whereas real identities such as $\sqrt{16} = 4$ are certified theorems derived from $\sqrt{a^2} = |a|$, stated and proved rather than computed.

- *Main-chain integration.* Integration is part of the claim : the module is compiled in every full build of the repository through its import by `Nods.lean`, so the formal content of this article is exercised continuously rather than archived once.

As elsewhere in the paper, no number-theoretic conjecture is touched by the module; the article's layout is inherited from an earlier preprint of the author and carries none of its original mathematical content.

8. Conclusion

The identification of the n -th root with a fractional power is not false; it is local. Over $\mathbb{C}^\times$ the symbol $x^{1/n}$ names one of n candidates without saying which : power and radical are not one operation wearing two names, but two structural roles — a quotient by μ_n and a section of the resulting n -fold cover — and the torsion subgroup μ_n is the exact measure of the difference, because removing it collapses the distinction : p_n becomes injective, and the radical reverts to being the inverse of the power, precisely when $\mu_n = \{1\}$ (Theorem 7). On $\mathbb{R}_+$ the collapse is then no miracle of real analysis : $\mu_n \cap \mathbb{R}_+ = \{1\}$, the cover has nothing to be nontrivial over, and the section is a function.

That the split is structural rather than rhetorical is the burden of the certified ladder. The quadratic solver exhibits both roots of $x^2 - bx - c = 0$ and factors the polynomial exactly when $\Delta \geq 0$ (Theorem 9); the Cardano construction exhibits a real root of the depressed cubic and, in its proof, is forced to meet the torsion split head-on, since v may not be chosen as an independent cube root of $-q/2 - s$ (Theorem 10); and the pure n -th root then flattens the ladder entirely, because one `exp / log` statement yields the unique positive root for every degree at once — the quintic root is not a further rung but the same theorem at $n = 5$ (Theorem 11).

The boundary of the claim is part of the claim. What dies at degree five is not the root but the mixed tower : field operations closed under roots of arbitrary degree solve every polynomial through degree four and fail at the general quintic, and Abel–Ruffini and Galois theory [3, 4] fix that boundary — cited here rather than proved, as they must be, since a formal proof is a separate programme, not a missing paragraph. In the same spirit the real arithmetic exhibited here is certified, not executed : integer radicals are computed by the proof system (`norm_num`), while statements such as $\sqrt{16} = 4$ are theorems about noncomputable constants, verified through $\sqrt{a^2} = |a|$; where the two regimes meet, the text states the boundary explicitly.

9. Further work

None of the following is done; each item marks a boundary of the present work rather than a claim.

- *The complex full picture.* In $\mathbb{C}^\times$ one should have $|\mu_n| = n$ and an n -valued root whose monodromy permutes the branches by μ_n , fusing the algebraic split of Section 4 with the

analytic picture behind the fractional-power notation. Cost : cyclotomic theory for $\mu_n(\mathbb{C}^\times)$ together with covering-space machinery.

- *The formal Galois wall*. Proving — rather than citing — Abel–Ruffini for the quintic inside Lean would make the boundary at degree five internal to the system. Cost : solvable-group theory, field extensions, and the insolubility of S_5 — a large, separate development.
- *Mixed-tower completions*. A translation step to the depressed form for the general cubic, and Ferrari’s quartic via its cubic resolvent, would close the certified solvable range through degree four. Cost : moderate, and the resolvent’s root choices are the torsion branch problems of Section 4 again.
- *Certified numerical radicals*. Interval methods with proof-carrying bounds would turn the certified but non-executable real radicals of Section 6 into verified enclosures — a bridge from statements to computation. Cost : an approximation layer for exp, log, and roots ; substantial, but routine relative to this development.

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